

Final Project — Data, Papers, and Replication Resources

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Final Project Resources

A curated, comprehensive list of datasets, replication packages, papers, and tools that students can use for the final project. Everything here is either freely accessible, accessible with a quick application, or distributed as part of an R package on CRAN. When a dataset requires credentialing or registration, that is noted explicitly.

The list is organized into:

1. Canonical causal-inference datasets (used in textbooks and the lectures)
2. Datasets bundled in R packages (zero-friction)
3. Public-use surveys and administrative microdata
4. Health, biomedical, and Mendelian-randomization data
5. Causal-discovery benchmark datasets
6. Replication archives and data repositories
7. Curated paper lists by lecture topic
8. Free books and online tutorials
9. Software ecosystem

A short closing section gives advice on choosing a project: the easiest path is to pick a published paper, get its replication package, redraw the DAG, and re-estimate using a method from the course.

1. Canonical causal-inference datasets

These are the datasets that appear repeatedly in the textbooks (HR, IR, Pearl, PJS) and in the course lectures and labs. Each one comes with a published causal question, so students do not have to invent one from scratch.

Dataset	Causal question	Where to get it	Lectures it touches
LaLonde NSW (Dehejia- Wahba subset)	Effect of job-training on earnings	R: <code>Matching::lalonge</code> , <code>causalsens::lalonge.exp</code> , <code>causalsens::lalonge.psid</code> ; CSV at https://users.nber.org/~rdehejia/data/.nswdata2.html	L5–L6 (estimation, ATT), L7 (sensitivity)

Dataset	Causal question	Where to get it	Lectures it touches
NHEFS (NHANES I Epidemiologic Follow-Up Study; HR companion data)	Effect of smoking cessation on weight gain	https://shared-resource.s3.amazonaws.com/wcm/causal-inference-book/nhefs.csv ; R: <code>causaldata::nhefs</code>	L4–L7 (the book’s running example)
RHC (Right Heart Catheteriza- tion, Connors et al. 1996)	Effect of swan-ganz catheter on 30-day mortality in ICU patients	R: <code>Hmisc::getHdata("rhc");</code> mirror at https://hbiostat.org/data/repo/rhc.html	L5–L6 (AIPW, doubly robust ATT)
Card-Krueger NJ-PA fast-food Cattaneo (2010) birth-weight	Effect of NJ minimum-wage hike on employment	https://davidcard.berkeley.edu/data_sets.html ; R: <code>wooldridge::njmin3</code>	L6 (DiD)
Angrist quarter-of-birth	Effect of maternal smoking on infant birth weight	R: <code>sensemakr::bw</code> , also bundled in Cattaneo’s Stata textbook companion	L5 (matching), L7 (sensitivity)
Card (1995) college proximity Acemoglu, Johnson & Robinson (2001) Project STAR	Returns to schooling using QoB as IV	R: <code>AER::CollegeDistance</code> for a parallel example; original at https://economics.mit.edu/people/faculty/josh-angrist/angrist-data-archive	L8 (IV)
	Returns to schooling using nearby-college as IV	R: <code>AER::CollegeDistance,</code> <code>wooldridge::card</code>	L8 (IV)
	Effect of institutions on GDP using settler mortality as IV	https://economics.mit.edu/people/faculty/daron-acemoglu/data-archive ; R: <code>AER::Acemoglu</code>	L8 (IV)
	Effect of small classes on test scores (Tennessee class-size RCT)	Harvard Dataverse: https://doi.org/10.7910/DVN/SIWH9F ; R: <code>AER::STAR</code>	L1 (RCT), L5
Oregon Health Insurance Experiment Sachs et al. protein signaling IHDP (Infant Health & Development Program)	Effect of Medicaid coverage on health and finance	https://www.nber.org/oregon (Finkelstein et al.)	L1 (RCT), L8 (IV; lottery → coverage)
	Discover signaling network from single-cell flow cytometry	R: <code>pcalg::sachs,</code> <code>bnlearn::sachs</code>	L9–L10 (discovery)
	Effect of intensive intervention on infant cognitive scores; standard ML-causal benchmark	Hill (2011) processed version: https://www.fredjo.com and <code>causaltoolbox</code>	L5–L6

Resources for these specific datasets

- Hernán & Robins (2025) *Causal Inference: What If* — full data + R/SAS/Stata/Python code at <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>
- Dehejia & Wahba processed LaLonde subsets — <https://users.nber.org/~rdehejia/data/.nswdata2.html>
- Cunningham *Mixtape* data: <https://github.com/scunning1975/mixtape>

2. Datasets bundled in R packages (zero-friction)

The fastest path to a project is to install one of these and pick a dataset from `data(package = "<name>")`. Every dataset listed has an associated published paper.

R package	What's inside
<code>causaldata</code>	NHEFS, Black-Smith, Castle, restaurant inspections, scorecard, mortgages — every dataset in <i>Causal Inference: The Mixtape</i> and <i>The Effect</i>
<code>causalsens</code>	LaLonde NSW + observational PSID controls
<code>Matching</code>	LaLonde, lung cancer, GerberGreenImai (turnout)
<code>MatchIt</code>	LaLonde
<code>AER</code> (Applied Econometrics with R)	CollegeDistance, CASchools, Fatalities, GrowthDJ, GunViolence, Cigarettes, GoldSilver — many natural-experiment classics
<code>wooldridge</code>	Every dataset from Wooldridge's textbooks (>100 econometric examples)
<code>Ecdat</code>	Economics datasets
<code>pcalg</code>	gmG, gmI, gmL — small simulated and real graphs for discovery
<code>bnlearn</code>	asia, alarm, hailfinder, sachs, child — Bayesian network benchmarks
<code>ipumsr</code>	Programmatic access to IPUMS extracts (CPS, ACS, IPUMS-International)
<code>tidycensus</code>	American Community Survey microdata via API
<code>nhanesA</code>	All public NHANES datasets, fetched on the fly
<code>survey</code>	Lumley's package — needed for any complex-sample design (NHANES, NHIS, etc.)
<code>WeightIt</code> , <code>cobalt</code> , <code>MatchThem</code>	Tools + small example datasets for IPW and matching

A “five-minute project starter”: `install.packages("causaldata"); library(causaldata); data(nhefs)` — and you have the running example from HR.

3. Public-use surveys and administrative microdata

These are large public-use sources that support an enormous variety of causal questions. All are free; some require a brief registration.

U.S. surveys

- **NHANES** — National Health and Nutrition Examination Survey, CDC. <https://wwwn.cdc.gov/nchs/nhanes/>. Continuous since 1999. Use the **nhanesA** R package.
- **NHIS** — National Health Interview Survey. <https://www.cdc.gov/nchs/nhis/>. Annual.
- **BRFSS** — Behavioral Risk Factor Surveillance System. <https://www.cdc.gov/brfss/>. State-level health behaviors.
- **NLSY97 / NLSY79** — National Longitudinal Survey of Youth. <https://www.bls.gov/nls/>. Earnings, education, family formation.
- **CPS** — Current Population Survey. <https://www.census.gov/programs-surveys/cps.html>. Use IPUMS-CPS for harmonized extracts.
- **ACS** — American Community Survey. <https://www.census.gov/programs-surveys/acs>. Use IPUMS-USA.
- **PSID** — Panel Study of Income Dynamics. <https://psidonline.isr.umich.edu/>. The longest household panel in the world.
- **HRS** — Health and Retirement Study. <https://hrs.isr.umich.edu/>. Aging, health, retirement.
- **MEPS** — Medical Expenditure Panel Survey. <https://meps.ahrq.gov/>. Healthcare cost and utilization.
- **SIPP** — Survey of Income and Program Participation. <https://www.census.gov/sipp/>.
- **ECLS-K** — Early Childhood Longitudinal Study. <https://nces.ed.gov/ecls/>.
- **Add Health** — National Longitudinal Study of Adolescent to Adult Health. <https://addhealth.cpc.unc.edu/>.

International / global

- **IPUMS International** — harmonized census microdata from 100+ countries. <https://international.ipums.org/>.
- **Demographic and Health Surveys (DHS)** — <https://dhsprogram.com/>. Free with registration.
- **World Bank Microdata Library** — <https://microdata.worldbank.org/>. LSMS, household surveys.
- **OECD Data** — <https://data.oecd.org/>.
- **WHO Global Health Observatory** — <https://www.who.int/data/gho>.
- **UNICEF MICS** — <https://mics.unicef.org/>.

Education

- **NCES** — National Center for Education Statistics: <https://nces.ed.gov/datatools/>.
- **Stanford Education Data Archive (SEDA)** — <https://edopportunity.org/get-the-data/seda-archive-downloads/>.

Crime, justice, environment

- **National Crime Victimization Survey (NCVS)** — via ICPSR.
- **EPA Air Quality System (AQS)** — <https://www.epa.gov/aqs>.
- **NOAA climate data** — <https://www.ncei.noaa.gov/>.

4. Health, biomedical, and Mendelian-randomization data

Public-use clinical / biomedical

- **MIMIC-III / MIMIC-IV** — de-identified ICU records, ~60K admissions. Free with a one-time CITI training and DUA. <https://physionet.org/content/mimiciv/>. Excellent for treatment-effect questions.
- **eICU Collaborative Research Database** — <https://physionet.org/content/eicu-crd/>.
- **NHLBI BioLINCC** — Framingham Heart Study, WHI, ARIC, etc. Apply with a 1-page protocol. <https://biolincc.nhlbi.nih.gov/home/>.
- **All of Us Research Program** — large NIH cohort. Researcher account required. <https://www.researchallofus.org/>.
- **dbGaP** — controlled-access genotype + phenotype. <https://www.ncbi.nlm.nih.gov/gap/>.
- **UK Biobank** — ~500K participants, deep phenotyping + genotyping. Application required, ~3 months. <https://www.ukbiobank.ac.uk/>.
- **TCGA / cBioPortal** — cancer genomics. <https://www.cbioportal.org/>.
- **Project Data Sphere** — historical cancer-trial control arms. <https://www.projectdatasphere.org/>.

Mendelian-randomization / causal genetic epidemiology

- **MR-Base / IEU OpenGWAS** — over 100K curated GWAS summary stats. <https://gwas.mrcieu.ac.uk/>. R packages: `TwoSampleMR`, `MendelianRandomization`.
- **GWAS Catalog (NHGRI-EBI)** — <https://www.ebi.ac.uk/gwas/>.
- **PheWAS Catalog** — <https://phewascatalog.org/>.

COVID-19 (still actively used as an applied source)

- **CDC COVID Data Tracker** — <https://covid.cdc.gov/covid-data-tracker/>.
- **JHU CSSE COVID-19 GitHub** — historical case/death series.
- **Our World in Data COVID** — <https://github.com/owid/covid-19-data>.

5. Causal-discovery benchmark datasets

For projects that learn the graph (PC, GES, FCI, NOTEARS, ...):

- **bnlearn benchmark networks** — `asia`, `alarm`, `hailfinder`, `child`, `insurance`, `mehra`, `pathfinder`, `win95pts`, `andes`. Ground-truth DAGs included. <https://www.bnlearn.com/bnrepository/>.
- **Sachs protein signaling** — `pcalg::sachs`. Eleven proteins, intervention experiments.
- **fMRI Netsim** — Smith et al. (2011) NeuroImage. Simulated fMRI time-series with known ground-truth connectivity. Useful if students want a structural-discovery angle on brain data.
- **CauseEffectPairs** — <https://webdav.tuebingen.mpg.de/cause-effect/>. 100+ real bivariate cause-effect pairs (used for two-variable discovery / additive-noise models).
- **CausalDiscoveryToolbox** (Python, but datasets work in R) — <https://github.com/FenTechSolutions/CausalDiscoveryToolbox>.
- **Tetrad** — Glymour-Spirtes-Scheines tool with bundled datasets. <https://www.cmu.edu/dietrich/philosophy/tetrad/>.
- **CausalBench** (Chevalley et al. 2022) — single-cell perturbation benchmarks for discovery. <https://github.com/causalbench/causalbench>.
- **ACIC Data Challenge** (Atlantic Causal Inference Conference) — 2016, 2017, 2018, 2019, 2022 challenges. Each gives ground-truth ATT on simulated-from-real data. <https://sites.google.com/view/acic2022/home>.

6. Replication archives and data repositories

The fastest way to build a serious project is to take a published paper, run its replication code as-is, then re-do the analysis under different causal assumptions.

Repository	What's there
AEA Data and Code Repository (openICPSR)	Mandatory replication packages for AER, AEJ, JEP, JEL papers since 2019. https://www.openicpsr.org/openicpsr/aea
ICPSR	Over 250K social-science datasets including replications. https://www.icpsr.umich.edu/
Harvard Dataverse	Most political-science and many epidemiology replication packages. https://dataverse.harvard.edu/
NBER Working Papers	Many include data/code links; especially good for IV and DiD applications. https://www.nber.org/papers
JASA / JRSS-B replication archives	Embedded in articles' supplementary materials
Code Ocean	Containerized reproducibility capsules. https://codeocean.com/explore
OSF (Open Science Framework)	Pre-registrations + data + code, particularly in psych and ed. https://osf.io/
Zenodo / Figshare	General-purpose research-data archives
The Replication Wiki (econ)	Catalog of econ replication studies. https://replication.uni-goettingen.de/
ReScience C / ReScience X	Computational-replication journal. https://rescience.github.io/

Causal-inference-specific replication corpora

- **Causal Inference: The Mixtape** — every figure and table reproducible. <https://github.com/scunning1975/mixtape>.
- **The Effect** (Huntington-Klein) — <https://github.com/NickCH-K/TheEffectAssistant>.
- **Hernán & Robins What If** book code — Python/R/SAS/Stata at <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>.
- **Cinelli & Hazlett sensemakr** — Darfur and other examples ship with the package.
- **EconML examples** (Microsoft) — <https://github.com/py-why/EconML>.
- **DoWhy examples** — <https://github.com/py-why/dowhy>.

7. Paper lists by lecture topic

A focused starting list, biased toward applied papers that are also good “redo with a different identification strategy” candidates. Many of these are already in the syllabus’s Applied Readings column; the entries below add depth.

Potential outcomes, RCTs, and partial identification (L1–L2)

- LaLonde (1986) *AER* — the canonical NSW analysis.

- Finkelstein et al. (2012) *QJE* — Oregon Medicaid lottery.
- Miguel & Kremer (2004) *Econometrica* — Kenya deworming RCT (interference).
- Duflo, Glennerster & Kremer (2007) *Handbook of Development Economics* — RCT design toolkit.
- Manski (1990) *AER* — non-parametric bounds on treatment effects.
- Manski & Pepper (2000) *Econometrica* — monotone IV applied to schooling.
- Kreider, Pepper, Gundersen & Jolliffe (2012) *AER* — bounds on SNAP and child health.
- Balke & Pearl (1997) *JASA* — bounds on IV models with non-compliance.

DAGs, identification, and good controls (L3–L4)

- Hernán, Hernández-Díaz & Robins (2004) *Epidemiology* — selection bias via DAGs.
- Greenland, Pearl & Robins (1999) *Epidemiology* — causal diagrams primer.
- Cinelli, Forney & Pearl (2022) *Sociological Methods & Research* — Crash Course in Good and Bad Controls.
- Elwert & Winship (2014) *Annual Review of Sociology* — collider bias.
- Shrier & Platt (2008) *BMC Medical Research Methodology* — short illustrated DAG primer.
- Pearl (1995) *Biometrika* — front-door (optional reading).

Estimation: regression, IPW, matching, AIPW (L5–L6)

- Rosenbaum & Rubin (1983) *Biometrika* — propensity score (foundational).
- Lunceford & Davidian (2004) *Statistics in Medicine* — IPW vs. stratification.
- Dehejia & Wahba (1999) *JASA* — matching on LaLonde.
- Abadie & Imbens (2011) *JBES* — bias-corrected matching.
- Glynn & Quinn (2010) *Political Analysis* — accessible AIPW tutorial.
- Robins, Hernán & Brumback (2000) *Epidemiology* — marginal structural models for HIV treatment.
- Chernozhukov et al. (2018) *Econometrics Journal* — double/debiased ML.
- Kang & Schafer (2007) *Statistical Science* — IPW vs. doubly-robust in practice.
- Stuart (2010) *Statistical Science* — matching review.

Difference-in-differences (L6 cont.)

- Card & Krueger (1994) *AER* — minimum wage in NJ.
- Bertrand, Duflo & Mullainathan (2004) *QJE* — DiD with serial correlation.
- Goodman-Bacon (2021) *Journal of Econometrics* — staggered DiD decomposition.
- Callaway & Sant’Anna (2021) *Journal of Econometrics* — staggered DiD estimator.
- Roth, Sant’Anna, Bilinski & Poe (2023) *Journal of Econometrics* — modern DiD review.

Sensitivity analysis (L7)

- Rosenbaum (2002) *Observational Studies* Ch. 4 — Γ -bounds.
- VanderWeele & Ding (2017) *Annals of Internal Medicine* — E-values.
- Cinelli & Hazlett (2020) *JRSS-B* — sensemakr (partial- R^2).
- Imbens (2003) *AER* — sensitivity in observational studies.
- Ding & VanderWeele (2016) *Epidemiology* — bounding-factor approach.

Instrumental variables (L8)

- Angrist & Krueger (1991) *QJE* — quarter-of-birth.
- Card (1995) — college proximity.
- Acemoglu, Johnson & Robinson (2001) *AER* — settler mortality.

- Imbens & Angrist (1994) *Econometrica* — LATE.
- Angrist, Imbens & Rubin (1996) *JASA* — IV with potential outcomes.
- Davey Smith & Hemani (2014) *Human Molecular Genetics* — Mendelian randomization.
- Bowden, Davey Smith & Burgess (2015) *International Journal of Epidemiology* — MR-Egger.
- Andrews, Stock & Sun (2019) *Annual Review of Economics* — weak-instrument review.

Causal discovery (L9–L10)

- Sachs et al. (2005) *Science* — protein-signaling networks.
- Maathuis, Kalisch & Bühlmann (2009) *Annals of Statistics* — IDA.
- Maathuis et al. (2010) *Nature Methods* — predicting effects in yeast.
- Colombo, Maathuis, Kalisch & Richardson (2012) *Annals of Statistics* — RFCI.
- Ramsey, Glymour, Sanchez-Romero & Glymour (2017) — high-dimensional GES.
- Spirtes, Meek & Richardson (1995) — FCI.
- Zhang (2008) *Artificial Intelligence* — augmented FCI for selection bias.
- Squires & Uhler (2023) *Annual Review of Statistics* — modern review.

Mediation (L11)

- VanderWeele (2016) *Annual Review of Public Health* — practitioner’s guide.
- Imai, Keele & Yamamoto (2010) *Statistical Science* — identification + sensitivity for mediation.
- Pearl (2001) *UAI* — natural direct/indirect effects.

8. Free books and online tutorials

Direct, freely accessible links — useful for filling gaps as the project progresses.

- **Hernán & Robins**, *Causal Inference: What If* — <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/> [the course’s primary applied reference]
 - **Peters, Janzing & Schölkopf**, *Elements of Causal Inference* — <https://library.oapen.org/handle/20.500.12657/26040>
 - **Cunningham**, *Causal Inference: The Mixtape* — <https://mixtape.scunning.com/>
 - **Huntington-Klein**, *The Effect* — <https://theeffectbook.net/>
 - **Pearl, Glymour & Jewell**, *Causal Inference in Statistics: A Primer* — sample chapters at <http://bayes.cs.ucla.edu/PRIMER/>
 - **Hsiang, Lieberman & Lobell**, *Causal Inference for the Brave and True* (Python) — <https://matheusfacure.github.io/python-causality-handbook/>
 - **MIT OCW 14.387 (Angrist)** — <https://ocw.mit.edu/courses/14-387-applied-econometrics-mostly-harmless-big-data-fall-2014/>
 - **Coursera “A Crash Course in Causality” (Roy)** — Penn.
 - **Anthology of Coding-Up Causal Estimators** — `causaldata` package vignettes mirror every chapter of *Mixtape* and *The Effect*.
 - **Microsoft DoWhy + EconML book chapters** — <https://www.pywhy.org/>
 - **Pearl’s blog (“Causal Analysis in Theory and Practice”)** — <http://causality.cs.ucla.edu/blog/>
 - **Andrew Heiss’s tutorials** (DAGs in `ggdag`, IPW, `sensemkr`) — <https://www.andrewheiss.com/research/>
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9. Software ecosystem (R-first, since the course requires R)

Modeling assumptions and identification

- `dagitty` — encode DAGs and ADMGs; built-in adjustment-set queries; web companion at <https://www.dagitty.net/>.
- `ggdag` — tidyverse-friendly DAG plotting on top of `dagitty`.
- `pcalg` — adjustment criterion (`gac`), generalized adjustment, IDA, PC, GES, FCI.

Estimation

- `WeightIt` + `cobalt` — modern IPW with balance diagnostics.
- `MatchIt` — matching with consistent diagnostics interface.
- `Matching` — Sekhon’s package; fast genetic matching.
- `survey` — Lumley’s complex-design tools (essential for NHANES, etc.).
- `sandwich` + `lmtest` — robust SEs.
- AIPW, `DoubleML` — doubly-robust + cross-fitting.
- `did`, `bacondecomp`, `fixest` — modern DiD.
- `ivreg`, `ivmodel`, `AER::ivreg` — IV.
- `MendelianRandomization`, `TwoSampleMR` — MR.

Sensitivity analysis (lecture 7)

- `sensemkr` — Cinelli–Hazlett partial- R^2 framework (with the Darfur example).
- `EValue` — VanderWeele–Ding E-values.
- `rbounds`, `sensitivitymv`, `sensitivitymw` — Rosenbaum Γ -bounds for matched studies.
- `tipr` — tipping-point / unmeasured-confounder framework.

Discovery

- `pcalg` — PC, GES, FCI, GIES, RFCI, IDA, generalized adjustment.
- `bnlearn` — wide collection of structure-learning algorithms; benchmark networks bundled.
- `causalDisco` — implementations of newer scores.
- Python complement: `causal-learn`, `dowhy`, `econml`, `causalnex`.

End-to-end “research package” frameworks

- `causaldata` — every dataset from the leading textbooks, no friction.
- `tidylog` + `targets` — for reproducibility of the analysis pipeline.
- `renv` — pin package versions so the reproducibility bundle actually runs.

How to choose a project (for groups still deciding)

A reliable recipe:

1. Pick a **published applied paper** from the lecture-by-lecture list above (Section 7).
2. Find its **replication package** in one of the archives in Section 6 (AEA, ICPSR, Harvard Dataverse).

3. Re-draw the **DAG** the paper *implicitly* used. This is almost always the most original part of a project — most applied papers leave their causal assumptions in prose.
4. State **identification** in your DAG using a method from the course (back-door / generalized adjustment / IV / DiD).
5. **Re-estimate** with at least one method from the course (a different one from the paper if you can — the comparison is usually the point).
6. Run a **diagnostic** (overlap plot, balance table, sensitivity analysis with **sensemakr** or an E-value).
7. Write up the seven-stage pipeline.

This is far easier than collecting your own data, and it gives you something to compare against. A “we re-analyzed paper X using method Y from the course and recovered / overturned its central estimate” project is often the strongest in the class.